Thammarat Yawisit (Pound)

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Education

University of Wisconsin–Madison
IceCube Summer Student Programme 2025 *May–Aug 2025*

- Grade: A (Overseas Training).

King Mongkut's Institute of Technology Ladkrabang (KMITL)
B.Eng. in Mechatronics and Automation Engineering *Aug 2022–Apr 2026*

- Grade: 3.19 / 4.00.
- Advisors: Assoc. Prof. Pittaya Pannil and Assoc. Prof. Amphawan Julsereewong.
- Thesis: *Design and Implementation of a Liveness-Aware Trigger System for High-Energy Neutrino Detection in Distributed Optical Arrays*.

Appointments and Experience

King Mongkut's Institute of Technology Ladkrabang (KMITL)
Research Engineer, School of Engineering *Oct 2026–present*

- Project: Installation and Commissioning of the SpinQ Gemini Nuclear Magnetic Resonance Quantum Computer (NMRCQ).

Chulabhorn Royal Academy – Princess Srisavangavadhana Faculty of Medicine
Research Engineer *Apr–Sep 2026*

- Project: Development and Validation of a Computational Framework for Radiation-Induced DNA Double-Strand Break Repair Kinetics.
- Supervisor: Associate Professor Thiansin Liamsuwan, Ph.D.

King Mongkut's Institute of Technology Ladkrabang (KMITL)
Research Assistant, School of Engineering *Sep 2025–Mar 2026*

- Project: Liveness-Aware Signal Processing and Trigger Development for Neutrino Telescopes.
- Supervisors: Prof. Pittaya Pannil and Assoc. Prof. Amphawan Julsereewong.

Wisconsin IceCube Particle Astrophysics Center (WIPAC)
Research Engineer Intern, IceCube Neutrino Observatory *May–Aug 2025*

- Project: Design and Validation of a Cosmic Muon-Based Testing Method for IceCube-Gen2 Optical Modules.
- Supervisors: Yuya Makino, Chris Wendt, and Albrecht Karle.

Thai Beverage Public Company Limited (ThaiBev)
Automation Engineer Intern *May–Jul 2024*

- Project: Process Automation Design and Predictive Maintenance for ThaiBev Fermentation Systems.
- Supervisors: Montri Yakul, Prasertsak Kongsap, Ittipon Singhaseni, and Inarm Satitanon.

Panasonic

Fellow of Panasonic Scholarship Asia 2023 *Oct 2023–Jul 2026*

- Supervisor: Thammarong Ondeesawat.

The Viriyah Insurance Public Company Limited

Data Analyst Intern *Mar–May 2022*

- Project: Insurance Claim System Visualization using Power BI.
- Supervisor: Warunya Naksuwan.

Selected Publications

Conferences

1. **Yawisit, T.**, and Pannil, P. (2026). *Mitigating Deadtime with Liveness-Aware Recursive Triggering in a Simulated Multi-PMT Optical Module for Neutrino Telescopes*. Oral presentation at the 28th International Conference on Computing in High Energy and Nuclear Physics (CHEP 2026), Track 2: Online & Real-Time Computing, Bangkok, Thailand, 28 May 2026 (Contribution 562). Proceedings submitted to *EPJ Web of Conferences*; under review. https://doi.org/10.48550/arXiv.2601.18114

2. **Yawisit, T.**, and Pannil, P. (2026). *Testing Geometric Scoring for a Liveness-Aware Recursive Trigger in a Simulated Multi-PMT Optical Module for Neutrino Telescopes*. Oral presentation at the 28th International Conference on Computing in High Energy and Nuclear Physics (CHEP 2026), Track 2: Online & Real-Time Computing, Bangkok, Thailand, 28 May 2026 (Contribution 565). Proceedings submitted to *EPJ Web of Conferences*; under review. https://doi.org/10.48550/arXiv.2601.17043

3. **Yawisit, T.**, and Pannil, P. (2026). *60-MSPS Coherent-State Measurement of Cherenkov Photon Responses with Contravariance Diagnostics for Astronomical Neutrino Detection*. Oral presentation in Plenary Session III: Physics & Engineering, International Nuclear Science and Technology Conference (INST 2026), Bangkok, Thailand, 13 July 2026.

4. **Yawisit, T.**, and Pannil, P. (2026). *Missing Light, Misleading Confidence: Optical-Module Readout Loss and Signal Reliability in Neutrino Telescopes*. Manuscript submitted to the 2027 IEEE Sensors Applications Symposium (SAS); decision pending. https://doi.org/10.48550/arXiv.2610.24003

Preprints, Technical Reports, and Research Outputs

5. **Yawisit, T.**, and Pannil, P. (2025; revised 2026). *Time-Aligned Multichannel Coherence Estimation under Intermittent Observations*. Preprint, arXiv:2512.20678. https://doi.org/10.48550/arXiv.2512.20678

6. **Yawisit, T.**, Makino, Y., and Wendt, C. (2025). *Design and Validation of a Cosmic Muon-Based Testing Method for IceCube-Gen2 Optical Modules*. Research technical report, IceCube Summer Student Programme 2025, Wisconsin IceCube Particle Astrophysics Center (WIPAC), University of Wisconsin–Madison. Presented at the WIPAC Group Meeting on 28 July 2025; uploaded to IceCube Wiki on 30 July 2025. Public version available via Zenodo. https://doi.org/10.5281/zenodo.18614095

7. **Yawisit, T.** (2025). *Construction and Cosmic Muon-Based Validation of Optical Modules for the IceCube-Gen2 Neutrino Observatory*. Overseas Training Technical Report, supplementary to the main research project of the IceCube Summer Student Programme 2025. https://doi.org/10.5281/zenodo.23020724

8. **Yawisit, T.** (2025). *The 2025 IceCube Summer Student Programme under the Royal Initiative of Her Royal Highness Princess Maha Chakri Sirindhorn*. Programme report indexed by the Information Technology Foundation under the Initiative of Her Royal Highness Princess Maha Chakri Sirindhorn. Institutional source: https://www.princess-it.org/publications/ https://doi.org/10.5281/zenodo.23021859

9. **Yawisit, T.**, Pattanawanichchakorn, P., Supachart, T., and Ukakimaphan, P. (2026). *A Reproducible Computational Framework for Integrated Kinematic Analysis of Serial Robotic Manipulators*. Research output, Zenodo. https://doi.org/10.5281/zenodo.19332410

Academic Training

Sino-Thai International Summer School on High Energy Physics and Astrophysics
Selected Participant; Thai-JUNO Consortium Registration Support *Jul 2026*
Sun Yat-sen University, Guangzhou, China

- Selected for the three-week programme covering high-energy and astroparticle physics, neutrino physics, particle detection, scientific computing, and data analysis.
- Programme topics included ROOT, Geant4, GLoBES, and cosmic-ray muon detection, with exposure to JUNO, LHAASO, CSNS, HIAF, and CiADS research directions.

Professional Service

28th International Conference on Computing in High Energy and Nuclear Physics (CHEP 2026)
Technical Reviewer, Conference Proceedings *2026*

- Assessed proceedings manuscripts for Track 2: Online & Real-Time Computing.

IEEE International Conference on Adaptive Intelligence, Modeling and Simulation (ICAIMS 2026)
Program Committee Member / Technical Reviewer *2026*

- Reviewed submissions in adaptive intelligence, modelling, and simulation.

Projects

P-CISE Bangkok Drainability Observatory (BDO)
Founder; Independent Engineering and Open-Data Project *Sep 2026–present*
Bangkok, Thailand

- Built and publicly deployed an independent drainage-monitoring prototype for Bangkok, applying automation and process-monitoring methods to urban sensing.
- Integrated a registry of 795 stations, including 582 with coordinates, with five source adapters, time-to-live caching, and database/static/live data merging.
- Implemented source-health and data-freshness classification, geographic and status filters, and handling of stale, failed, unknown, and static-only records.
- Deployed the application on Streamlit Community Cloud; developing the research manuscript and further analysis.

Live project: https://bangkok-drainability-observatory.streamlit.app/

Teaching and Leadership

Electromechanical Laboratory, Automation Engineering, KMITL
Laboratory Teaching Assistant *Nov 2025–Mar 2026*

- Prepared electrical wiring and control-panel equipment, supported second-year laboratory sessions, and supervised wiring standards, safety procedures, and experimental workflows.
- Supervisor: Assoc. Prof. Pittaya Pannil.

Process Automation Laboratory, Automation Engineering, KMITL
Laboratory Teaching Assistant *Nov 2024–Mar 2025*

- Supported Process Automation and Microcontroller Systems sessions, including hardware/software integration, process-flow monitoring, plant monitoring, and reproducible laboratory setup.
- Supervisor: Asst. Prof. Krit Smerpitak.

Mechatronics and Automation Engineering, KMITL
President and Secretary General; Student Mentoring *Mar 2023–Mar 2026*

- Mentored undergraduates on academic planning, research opportunities, project development, and engineering career pathways.
- Coordinated student activities and communication with faculty.
- Additional leadership included departmental student-committee service and Vice-President of the 61st KMITL Engineering student body.

Selected Invited Talks and Seminars

Sino-Thai International Summer School on HEP and Astrophysics 2026
Invited Speaker in Student Session; Sun Yat-sen University, Guangzhou, China *Jul 2026*
Construction and Cosmic Rays Muon-Based Optical Modules Validation for IceCube-Gen2 Neutrino Observatory.

Siam Physics Congress 2026
Invited Speaker *May 2026*
Design and Validation of a Cosmic Muon-Based Testing Method for IceCube-Gen2 Optical Modules.

Fundamental Particle Physics Training 2026, Chulalongkorn University
Invited Speaker; Thai-CERN / DESY / GSI-FAIR / IceCube programme *Apr 2026*
Design and Validation of a Cosmic Muon-Based Testing Method for IceCube-Gen2 Optical Modules.

KMITL Overseas Training Research Seminar
Research Presentation *Oct 2025*
Construction and Cosmic Muon-Based Module Validation of Optical Modules for the IceCube-Gen2 Neutrino Observatory.

Thai-IceCube Collaboration Seminar 4, Chiang Mai University
Research Presentation *Oct 2025*
IceCube Summer Research on Construction and Cosmic Muon-Based Validation for IceCube-Gen2 Optical Modules.

WIPAC Group Meeting, University of Wisconsin–Madison
Research Presentation *Jul 2025*
Design and Validation of a Cosmic Muon-Based Testing Method for IceCube-Gen2 Optical Modules.

Thai-IceCube Collaboration Official Visit and MoA-Signing Programme
Experience-Sharing Presentation *Jun 2025*
IceCube Summer School 2025 Experience Sharing.

Industrial Internship Seminar, KMITL
Seminar Speaker *May 2025*
The Roles of Automation Engineers in the Food and Beverage Industry.

Academic Planning Seminar, KMITL
Invited Speaker *Mar 2025*
Strategic Planning of Study Path and Curriculum Management for Undergraduates.

Programme Introduction Seminar, KMITL
Seminar Speaker *Feb 2025*
Overview of Industrial Automation Based on ISA-95 Standard.

Engineering Admission Interview Seminar, KMITL
Seminar Speaker *Jan 2025*
Perspectives on Automation Engineering Program: Trends and Global Extensions.

Internship Experience Seminar, KMITL
Seminar Speaker *Oct 2024*
Pathway and Overview of Earlier Internship in Automation Engineering.

Thai Beverage Public Company Limited (ThaiBev)
Internship Technical Presentation *Jul 2024*
Integrated Industrial Process Automation, Computational Simulation, and Sustainable Energy Management: A Case Study from ThaiBev.

Scholarships and Research Support

Thai-JUNO Consortium Scholarship *2026*
Registration support for the Sino-Thai International Summer School on High Energy Physics and Astrophysics, Sun Yat-sen University, Guangzhou, China.

CHEP 2026 Student Financial Support *2026*
Full registration waiver for CHEP 2026, Chulalongkorn University, Bangkok, Thailand.

IceCube Summer Research Fellowship *2025*
Institutional research-training support from WIPAC, University of Wisconsin–Madison, United States.

Royal Patronage Full Scholarship, IceCube Summer Student Programme *2025*
Supported by the Information Technology Foundation under the Initiative of Her Royal Highness Princess Maha Chakri Sirindhorn.

KMITL Honor Institutional Prestige Scholarship *2025*
One-year full tuition waiver from the School of Engineering, KMITL.

KMITL Top-Ranked Scholarship for Overseas Training in North America *2025*
Institutional support for international research training from the Office of International Affairs, KMITL.

Panasonic Scholarship Asia / Thailand Future Gear Award *2023*
Long-term scholarship support covering tuition, academic development, leadership training, and corporate social engagement.

Royal Thai Air Force National "Chang Phueak" Programme *2022*
Selected through a competitive national scholarship programme; ranked 13th of 1,242 applicants.

Awards and Honours

- **Royal Consideration for the IceCube Summer Student Programme (2025).** Selected to represent Thailand; first Thai engineering student selected for the programme.
- **KMITL Honor Institutional Prestige Award (2025).**
- **KMITL Pioneering Achievement Award (2025).**
- **ThaiBev Internship Special Contribution Award (2024).**
- **Panasonic Scholarship Asia: Founder's Philosophy Award (2023).**

Languages

Thai: Native.
English: IELTS Academic Overall 6.5 (6 December 2025).
Listening: 6.5 Speaking: 6.5 Reading: 6.0 Writing: 6.0

Professional Memberships

American Physical Society (APS)	2025–present
Institute of Electrical and Electronics Engineers (IEEE)	2025–present
International Society of Automation (ISA)	2024–present

References

Professor Dr. Surin Khomfoi
Vice President, Office of the National Higher Education, Science, Research and Innovation Policy Council (NXPO)
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Dean, School of Engineering,
King Mongkut's Institute of Technology Ladkrabang
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Associate Professor Dr. Pittaya Pannil
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King Mongkut's Institute of Technology Ladkrabang
Academic Advisor pittaya.pa@kmitl.ac.th

Associate Professor Thiansin Liamsuwan, Ph.D.
Assistant Dean for Graduate Studies,
Princess Srisavangavadhana College of Medicine, Chulabhorn Royal Academy
Research Supervisor thiansin.lia@cra.ac.th

Dr. Yuya Makino
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University of Wisconsin–Madison
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